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Small polaron hopping conduction in V_2O_5 –Sb– TeO_2 glasses

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Abstract

The dc electrical conductivity of glasses in the system V_2O_5 –Sb– TeO_2 prepared by press quenching was studied at temperatures between 303 and 473 K. The composition range of the glass formation region was found to be $10 \leq TeO_2 \leq 100$ mol%, $0 \leq V_2O_5 \leq 70$ mol%, and $0 \leq Sb \leq 20$ mol%, respectively. The glasses indicated n-type semiconductors from the measurement of thermoelectric power. The dc conductivities at 473 K for the present glasses were determined to be 6.38×10^{-6} – 7.13×10^{-3} S cm⁻¹, indicating that the conductivity increased with increasing V_2O_5 concentration. Sb content also contributed to increase the conductivity and decrease activation energy for electrical conduction. A model of redox reaction during melting was proposed and quantitatively explained the reaction between V_2O_5 and Sb. A glass of composition $70V_2O_5 \cdot 20Sb \cdot 10TeO_2$ (mol%) having a conductivity of 7.13×10^{-3} S cm⁻¹ at 473 K was found to be the highest conductive glass among the previous vanadium–tellurite glasses. From the conductivity–temperature relation, it was found that small polaron hopping model was applicable at the temperature above $\frac{1}{2}\Theta_D$ (Θ_D : the Debye temperature); the electrical conduction at $T > \frac{1}{2}\Theta_D$ was due to adiabatic small polaron hopping of electrons between vanadium ions for $V_2O_5 \geq 50$ mol%, and non-adiabatic for $30 \leq V_2O_5 < 50$ mol%. The polaron bandwidth ranged from 0.052 to 0.148 eV in the adiabatic region, and the value was < 0.030 eV in the non-adiabatic region. The hopping carrier mobility varied from 2.65×10^{-6} to 2.90×10^{-4} cm² V⁻¹ s⁻¹ at 473 K. The carrier density was obtained to be of the order of 10^{20} – 10^{21} cm⁻³, and temperature dependence of the carrier density was barely present between 423 and 473 K. The conductivity of the present glasses was primarily determined by hopping carrier mobility. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

Electrical-conductive glasses containing transition metal oxide (TMO) [1–24] have been widely studied for their electrical-conduction mechanism. Generally, the direct current (dc) conduction mechanism at high temperature ($T > \frac{1}{2}\Theta_D$, T is

the absolute temperature and Θ_D is the Debye temperature, respectively) was believed to occur by small polaron hopping (SPH) model based on a strong electron–lattice interaction [1–16]. On the other hand, at low temperature (below about RT, RT is room temperature), the electrical conduction was interpreted by variable-range hopping (VRH) [15–18]. Recently, for V_2O_5 – Bi_2O_3 glasses doped with $BaTiO_3$ [19] and V_2O_5 – CoO – TeO_2 glasses containing microcrystalline clusters [20], the conductions were investigated by using Schnakenberg’s model [25] and multiphonon tunneling model proposed by Shimakawa [26], respectively.

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In addition, technological applications of electrical-conductive glasses were also studied. Particularly, electrical applications of highly conductive tellurite glasses containing TMO were investigated [27–31], and the results of these investigations suggested that the highly conductive glasses have a potential applicability as electrical devices (e.g., memory switching [27] and gas sensor [28–31]).

With the above background, we have prepared highly conductive tellurite glasses containing TMO, such as V_2O_5 –R– TeO_2 (R = Sb_2O_3 [1], SnO [2], Bi_2O_3 [3] or ZnO [4]), and studied their electrical properties. Among these glasses, the highest conductivity was obtained to be $1.46 \times 10^{-3} \text{ S cm}^{-1}$ at 473 K for $62V_2O_5 \cdot 4Sb_2O_3 \cdot 33TeO_2$ (mol%) glass [1] (oxygen gas-sensing behavior was confirmed for this glass [30]). The electrical conduction of these tellurite glasses was attributed to adiabatic SPH for $V_2O_5 \geq 50 \text{ mol\%}$, and non-adiabatic SPH for $V_2O_5 < 50 \text{ mol\%}$ [1–4]. From the obtained results [1–4], we found that the conductivity of these ternary vanadium–tellurite glasses depended on the cation radius of the third component (Sb_2O_3 , SnO , Bi_2O_3 or ZnO); the glasses containing cations with the lower ionic radius of $Sb^{5+} = 0.061 \text{ nm}$ gave a higher conductivity than those glasses containing $Sn^{2+} = 0.069 \text{ nm}$, $Bi^{5+} = 0.075 \text{ nm}$ or $Zn^{2+} = 0.075 \text{ nm}$ [32]. For these ternary glasses, it was also found that oxygen (O) ions of the third component were an obstacle to the hopping of electrons between vanadium (V) ions, e.g., $V-O-Zn-O-V$ [4]. From this, it was expected that highly conductive vanadium–tellurite glasses would result from substitution of a metal element as the third component for an oxide.

The purposes of the present study were three-fold. The first was to prepare ternary vanadium–tellurite glasses with electrical conductivity higher than those of the previous glasses [1–4], and than those of V_2O_5 – Sb_2O_3 – TeO_2 glasses [1] in particular. In the present study, antimony (Sb) with the smallest ionic radius among the above cations was selected as the third component. The second was to investigate the effect of Sb concentration on the electrical conductivity. The third was to clarify the mechanism of electrical conduction for V_2O_5 –Sb– TeO_2 glasses and to discuss the V_2O_5 -concentration dependence of the conduction.

2. Experimental procedure

Reagent-grade raw materials (V_2O_5 : 99.9%, Sb : 99.9% and TeO_2 : 99.99%) were mixed in an argon atmosphere in prescribed compositions, and a batch (5 g) was melted in an argon atmosphere in a porcelain crucible in an electric furnace at 750°C for 1 h. The melt was then poured on to a copper plate block, and quenched by pressing with another copper plate block. Thus, 1 mm thick glass plates of 4 cm^2 and 5 g in weight were obtained. The amorphous nature of the samples was checked by X-ray diffraction (XRD) analysis. The glass transition temperature (T_g) of the amorphous samples was measured by differential thermal analysis (DTA) at a heating rate of $10^\circ\text{C min}^{-1}$ (accuracy in the measurement of T_g was $\pm 2 \text{ K}$). The glass formation region of the present glasses was determined by using the results of XRD analysis.

The concentration of V^{4+} ion and the total vanadium ion concentration (defined as $C_v = V^{4+}/V_{\text{total}}$) were determined by titration using $KMnO_4$ and $FeSO_4(NH_4)_2SO_4$ solutions. The density of the glasses was measured at 293 K by a Gay-Lussac pycnometer using toluene. For the samples of the same composition from different batches, the agreement of the C_v values and the measured densities indicated to be within 9% and 1%, respectively.

The four-point probe method was employed to measure the dc electrical resistivity in argon atmosphere from 303 K to 473 K. Electrodes were prepared with silver paste. A dc current of $1 \mu\text{A}$ was applied to each sample for the resistivity measurements. The values of dc electrical conductivity on the same samples in different runs agreed within 4%, and samples of the same composition from different batches gave agreements within 5% for room-temperature conductivity.

3. Results

No differences between nominal and analytical concentration of V_2O_5 were found (see Table 1), hence analyzed concentration was used to describe the glass composition. The XRD of these

Table 1
Chemical composition and physical properties of V_2O_5 –Sb– TeO_2 glasses^a

Sample no.	Nominal composition (mol%)		$T^b \pm 1$ (K)	$W \pm 0.004$ (eV) (423–473K)	$d \pm 0.0400$ (g cm ⁻³)	$N \pm 0.0200$ ($\times 10^{22}$ cm ⁻³)	$R \pm 0.02$ (nm)	$\theta_D \pm 1$ (K)	$v_o \pm 0.01$ ($\times 10^{13}$ s ⁻¹)
	V_2O_5	Sb	TeO_2						
1	70[68.5] ^c	5	25	0.403	3.7188	1.8081	0.381	776	1.62
2	70[71.2]	10	20	0.378	3.5043	1.7226	0.372	716	1.49
3	70[69.0]	20	10	0.317	3.3679	1.6932	0.389	666	1.39
4	60[60.9]	20	20	0.376	3.6720	1.6035	0.396	686	1.43
5	50[50.9]	20	30	0.413	3.8923	1.4356	0.411	706	1.47
6	40[38.9]	20	40	0.443	4.3005	1.2864	0.426	786	1.64
7	30[30.5]	20	50	0.465	4.4139	1.0039	0.463	806	1.68
8	20[18.2]	20	60	0.412	4.6173	0.7094	0.520	806	1.68

^a d , density; W , activation energy for electrical conduction; N , vanadium ion density, R , mean V-ion spacing.

^b T , temperature at which the onset of the change of the slopes (in Fig. 2) was confirmed.

^c Analyzed concentration of V_2O_5 .

quenched melts for various compositions showed a broad peak, which characterized amorphous nature of the materials.

Fig. 1 shows the glass formation region of the system V_2O_5 –Sb– TeO_2 . The composition range of the glass formation region was found to be $10 \leq TeO_2 \leq 100$ mol%, $0 \leq V_2O_5 \leq 70$ mol%, and $0 \leq Sb \leq 20$ mol%, respectively. The glass transition temperatures were $T_g = 489$, 517 and 546 K for $70V_2O_5 \cdot 30TeO_2$, $70V_2O_5 \cdot 10Sb \cdot 20TeO_2$ and $70V_2O_5 \cdot 20Sb \cdot 20TeO_2$ glasses, respectively. T_g increased with increasing Sb concentration.

The glasses appeared to be n-type semiconductors from the measurement of thermoelectric power. No dc polarization due to ionic current was observed for all these glasses.

Fig. 2 shows the temperature dependence of the dc electrical conductivity, σ , for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 20 - 70$). This relationship between $\log(\sigma T)$ and T^{-1} is described as follows:

$$\sigma = (\sigma_0/T) \exp(-W/kT), \quad (1)$$

where σ_0 is the pre-exponential factor, W the activation energy for electrical conduction and k is the Boltzmann constant. DC conductivity increased with increasing V_2O_5 concentration and decreased with decreasing temperature as shown in Fig. 2. Two slopes were observed for

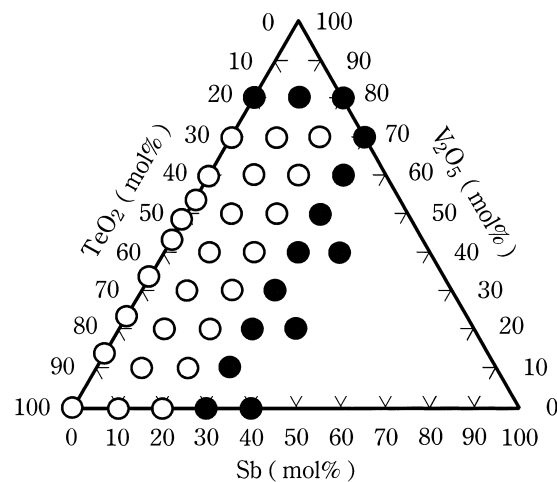


Fig. 1. Glass formation region of the system V_2O_5 –Sb– TeO_2 glasses. (○) glass; (●) crystallized.

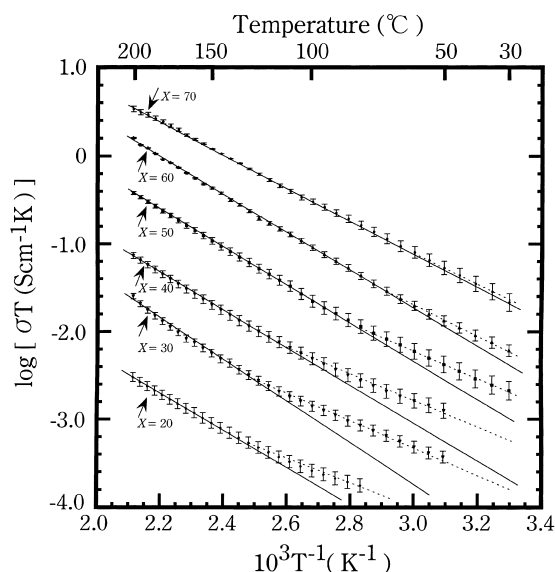


Fig. 2. Temperature dependence of dc conductivity for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%). The solid lines are calculated by using the least-square technique at $T \gtrsim \theta_D/2$. The dotted lines are drawn as a guide for the eye.

$XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 20 - 70$), suggesting that the conductivities were affected by two kinds of electrical conduction. Thus, in order to determine the temperature at which the onset of the change of these slopes was confirmed, we fitted the data in Fig. 2 to the relation $\ln(\sigma T) \propto T^{-1}$ by the least-square technique between 423 and 473 K. The solid lines were drawn by this technique as shown in Fig. 2 and the temperatures are given in Table 1, together with the temperatures of $70V_2O_5 \cdot YSb \cdot (30 - Y)TeO_2$ glasses (mol%) ($Y = 5, 10$ and 20). From the regression analysis, $r^2 = 0.9900 - 0.9997$ (r is the correlation coefficient) and $\text{Range} = 1.00 \times 10^{-3} \text{ S cm}^{-1}$ in σ were obtained, which indicate a satisfactory fit at temperatures between 423 and 473 K. The activation energy, W , in this temperature range was calculated from the $\ln(\sigma T) - T^{-1}$ relation, and the values are also given in Table 1. (The change of the slopes in the intermediate temperature region, which were drawn by the dotted lines in Fig. 2, will be discussed in Section 4.6.)

The σ values at 473 K were determined to be $6.38 \times 10^{-6} - 7.13 \times 10^{-3} \text{ S cm}^{-1}$ for the present

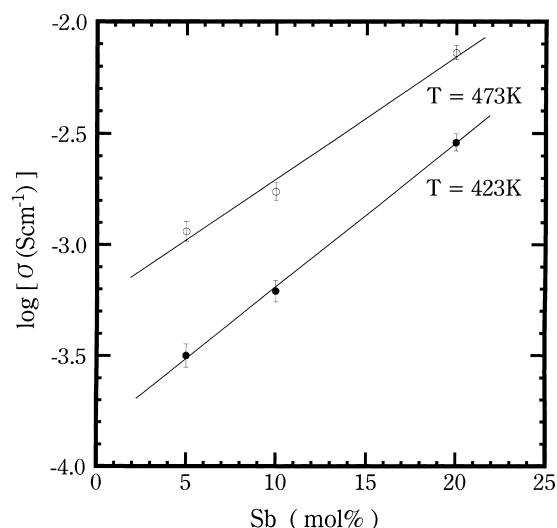


Fig. 3. Effect of Sb concentration on dc conductivity. The lines are drawn as a guide for the eye.

$XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 20 - 70$). For $X = 60$ and 70 , the σ values at 473 K were higher than that of $62V_2O_5 \cdot 4Sb_2O_3 \cdot 33TeO_2$ glass (mol%) ($\sigma = 1.46 \times 10^{-3} \text{ S cm}^{-1}$ at the same temperature [1]). Thus, the highly conductive glasses were obtained by substitution of Sb for Sb_2O_3 in the V_2O_5 – TeO_2 system.

Fig. 3 shows the effect of Sb concentration on dc conductivity for $70V_2O_5 \cdot YSb \cdot (30 - Y)TeO_2$ glasses (mol%) ($Y = 5, 10$, and 20). The dc conductivity at 423 and 473 K increased with increasing Sb concentration, respectively, which indicated that the increase of Sb concentration caused an increase of σ . The effect of Sb concentration on σ will be discussed in Section 4.6.

4. Discussion

4.1. Oxygen molar volume, V_O^*

The glass formation region of the present glasses was found to be larger than that of V_2O_5 – Sb_2O_3 – TeO_2 glasses ($30 < TeO_2 < 100 \text{ mol\%}$, $0 < V_2O_5 < 70 \text{ mol\%}$ and $0 < Sb_2O_3 < 12 \text{ mol\%}$ [1]). This result is due to the fact that the present glasses were thermally stable in

comparison with the thermal stability of V_2O_5 – Sb_2O_3 – TeO_2 glasses, because the T_g values of V_2O_5 – Sb – TeO_2 glasses ($T_g = 517$ and 546 K for $70V_2O_5 \cdot 10Sb \cdot 20TeO_2$ and $70V_2O_5 \cdot 20Sb \cdot 20TeO_2$ glasses, respectively) were higher than those of V_2O_5 – Sb_2O_3 – TeO_2 glasses ($T_g = 510$ – 516 K for $V_2O_5 = 62$ – 50 mol%).

Thermal stability of glasses is a result of the glass structure. Generally, thermally stable glasses have a closely packed structure [14,33–36]. Inversely, the structure of thermally unstable glasses possesses a loose packing [14,33–36]. Thus, we assumed the present glasses to have a more closely packed structure. For tellurite glasses, when a layered-network structure including a TeO_3 trigonal pyramid unit increases in concentration, the glasses form a closely packed structure [34,36], while the glasses form a loosely packed structure, when the network structure with a TeO_4 tetragonal pyramid unit increases in concentration [34,36]. Since the present glasses became thermally stable by containing Sb substituted as Sb_2O_3 , we assumed that the glass structure containing the TeO_4 units changed to the structure with the TeO_3 units by releasing oxygen: $TeO_4 \rightarrow TeO_3 + 1/2O_2$.

The differences between these structures can be compared in terms of the oxygen molar volume, V_O^* [14,33–36]. When the TeO_4 unit in the glass increases, V_O^* becomes larger than that for the glasses formed by the structure with the TeO_3 unit; generally, $V_O^* > \text{about } 13 \text{ cm}^3 \text{ mol}^{-1}$ [14,33–36]. So, we discuss the differences between the two glass structures for V_2O_5 – Sb – TeO_2 and V_2O_5 – Sb_2O_3 – TeO_2 glasses from the view point of

V_O^* . The V_O^* value of the present glasses is calculated by using the following equation.

$$V_O^* = \frac{(MW_{V_2O_5} - 16C_v)x + MW_{TeO_2}y}{d[(5 - C_v)x + 2y]}, \quad (2)$$

where $MW_{V_2O_5}$ is the molecular weight of V_2O_5 , MW_{TeO_2} the molecular weight of TeO_2 , x and y the mole fractions of V_2O_5 and TeO_2 , respectively, d the density of the glasses, and C_v is the fraction of reduced vanadium ion. The d and C_v values are summarized in Tables 1 and 2, respectively.

Fig. 4 shows the effect of V_2O_5 concentration on V_O^* for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 30$ – 70). V_O^* was obtained to be $V_O^* = 12.7$ – $11.7 \text{ cm}^3 \text{ mol}^{-1}$ for $V_2O_5 = 30$ – 70 mol%, lesser than those of the tellurite glasses reported previously [1–4,33–35]. From this, these V_O^* values mean that the TeO_3 unit in the present glasses increases compared with that in the V_2O_5 – Sb_2O_3 – TeO_2 glasses ($V_O^* = 14.5$ – $13.1 \text{ cm}^3 \text{ mol}^{-1}$ for $V_2O_5 = 11$ – 62 mol% [1]) and that the present glasses form the closely packed structure as assumed above by changing from the TeO_4 to the TeO_3 unit ($TeO_4 \rightarrow TeO_3 + 1/2O_2$), resulting in the present glasses becoming thermally stable. In addition, comparing the measured T_g of V_2O_5 – Sb – TeO_2 glasses (see Section 3) with those of V_2O_5 – Sb_2O_3 – TeO_2 glasses [1], we also confirmed that the results of T_g supported the V_O^* values of the present glasses. Theoretical V_O^* values calculated using theoretical C_v values are also drawn in Fig. 4 (theoretical C_v values were calculated using different redox models in Eqs. (10a) and (10b), and Eqs. (11a) and (11b)). Comparison

Table 2
Polaron hopping parameters of V_2O_5 – Sb – TeO_2 glasses

Sample no.	C_v^a	$W_H \pm 0.004$ (eV) (423–473 K)	$J \pm 0.004$ (eV) (423–473 K)	$r_p \pm 0.001$ (nm)	$e_p \pm 0.03$	$N(E_F)$ ($\text{eV}^{-1} \text{ cm}^{-3}$) (423–473 K)	$\gamma_p \pm 0.1$
1	0.078 ± 0.005	0.388	0.062	0.154	2.53	$5.80 \pm 0.40 \times 10^{19}$	11.5
2	0.144 ± 0.013	0.363	0.087	0.156	2.62	$1.15 \pm 0.20 \times 10^{20}$	11.8
3	0.210 ± 0.019	0.302	0.148	0.157	3.12	$5.94 \pm 0.90 \times 10^{20}$	10.6
4	0.217 ± 0.020	0.361	0.089	0.160	2.63	$1.48 \pm 0.20 \times 10^{20}$	12.0
5	0.229 ± 0.021	0.398	0.052	0.166	2.28	$2.96 \pm 0.20 \times 10^{19}$	13.0
6	0.349 ± 0.031	0.428	0.022	0.172	2.42	$3.41 \pm 0.70 \times 10^{18}$	12.6
7	0.480 ± 0.043	0.450	0.020	0.187	1.80	$2.32 \pm 0.80 \times 10^{18}$	12.8

^a C_v : fraction of reduced vanadium ion ($= V^{4+}/V_{\text{total}}$).

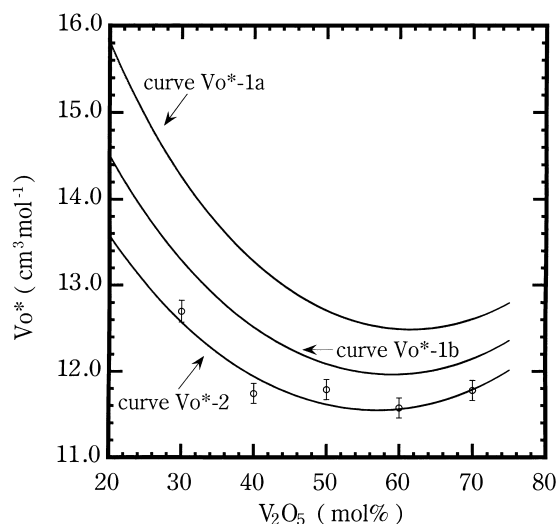


Fig. 4. Effect of V_2O_5 concentration on oxygen ion molar volume, V_o^* , for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%). The curves $V_o^*-1(a)$ and (b) are calculated using Eq. (2) with theoretical values of C_v (curves $C_v-1(a)$ and (b) in Fig. 9) and d (Fig. 10). The curve V_o^*-2 is calculated using Eq. (2) with the curve C_v-2 in Fig. 9 and the d value.

of theoretical and experimental C_v values will be discussed in Section 4.4.

4.2. Relationship between activation energy for conduction and mean V-ion spacing

The activation energies for electrical conduction, W , at temperatures between 423 and 473 K were found to be $W = 0.317\text{--}0.465$ eV (Table 1) for the present glasses. In our previous reports, we obtained $W = 0.38\text{--}0.50$ eV for $V_2O_5\text{--}Sb_2O_3\text{--}TeO_2$ glasses [1], $W = 0.38\text{--}0.44$ eV for $V_2O_5\text{--}SnO\text{--}TeO_2$ glasses [2], $W = 0.40\text{--}0.53$ eV for $V_2O_5\text{--}Bi_2O_3\text{--}TeO_2$ glasses [3] and $W = 0.39\text{--}0.50$ eV for $V_2O_5\text{--}ZnO\text{--}TeO_2$ glasses [4]. The W values of the present glasses nearly agreed with these glasses. Further, W depended on the mean V-ion spacing, R , for the glasses containing fixed concentrations of the third component, such as $XV_2O_5 \cdot 10SnO \cdot (90 - X)TeO_2$ (mol%) ($X = 20\text{--}60$), $XV_2O_5 \cdot 20SnO \cdot (80 - X)TeO_2$ (mol%) ($X = 20\text{--}50$) [2], $XV_2O_5 \cdot 10ZnO \cdot (90 - X)TeO_2$ (mol%) ($X = 30\text{--}60$), $XV_2O_5 \cdot 6Bi_2O_3 \cdot (94 - X)TeO_2$ glasses (mol%) ($X = 30\text{--}69$) [3] and $XV_2O_5 \cdot 20ZnO \cdot (80 - X)$

TeO_2 (mol%) ($X = 30\text{--}60$) [4]. These results suggested that the electrical conduction was due to small polaron hopping (SPH) between V-ions [1–4]. Thus, in the present study, the relationship between W and R should be confirmed for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 20\text{--}70$).

In order to confirm the W – R relation, the V-ion density, N , was calculated using the following formula:

$$N = 2[(d \cdot W_{V_2O_5} / MW_{V_2O_5}) N_A], \quad (3)$$

where d is the density, $W_{V_2O_5}$ the weight fraction of V_2O_5 , $MW_{V_2O_5}$ the molecular weight of V_2O_5 and N_A is the Avogadro number, respectively. The relationship between N and R is generally described as follows:

$$R = (1/N)^{1/3}. \quad (4)$$

The calculated values of N and R are summarized in Table 1. Fig. 5 shows the relationship between W and R for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 20\text{--}70$). The W values increased with increasing R , depending on R for $X = 30\text{--}70$ mol%.

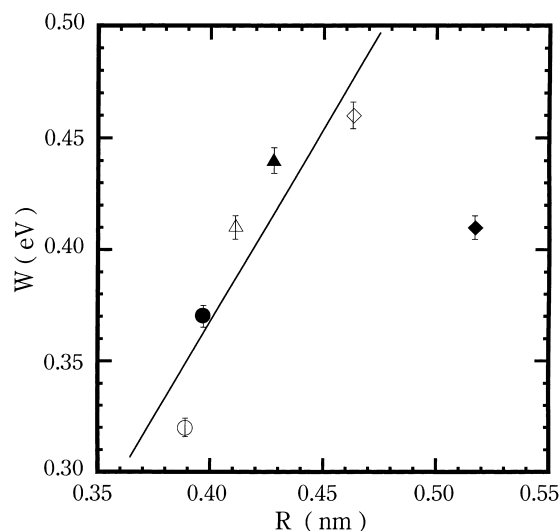


Fig. 5. Relationship between activation energy, W , and V-ion spacing, R , for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%). ($\circ$) $X = 70$, ($\bullet$) $X = 60$, ($\triangle$) $X = 50$, ($\blacktriangle$) $X = 40$, ($\diamond$) $X = 30$, ($\blacklozenge$) $X = 20$. The lines are drawn by using the least-square technique for the R values between 0.389 and 0.463 nm.

This result was similar to those for $V_2O_5 - R - TeO_2$ ($R = Sb_2O_3$ [1], SnO [2], Bi_2O_3 [3] or ZnO [4]), suggesting that the SPH occurred between V-ions.

For SPH conduction, the W - R relation is expressed as follows [37]:

$$W = W_0 + a(R - R_0). \quad (5)$$

In Fig. 5, the W - R relation nearly satisfies Eq. (5) for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 30$ – 70), and a was obtained to be 1.7 eV nm^{-1} for the R values between 0.389 and 0.463 nm . The a value was almost the same as $a = 1.4 \text{ eV nm}^{-1}$ for V_2O_5 - MnO - TeO_2 glasses [21]. On the other hand, Hirashima et al. [11] found that the W values of Fe_2O_3 - V_2O_5 - P_2O_5 glasses with high Fe_2O_3 content (more than $25 \text{ mol\%}$) were lower below those of the glasses with lower Fe_2O_3 content, (e.g., $W = 0.53 \text{ eV}$ for $10Fe_2O_3 \cdot 60V_2O_5 \cdot 30P_2O_5$ glass and $W = 0.40 \text{ eV}$ for $25Fe_2O_3 \cdot 60V_2O_5 \cdot 15P_2O_5$ glass (mol%)), and suggested a hopping between V and Fe ions in addition to that between V ions [11]. However, no such above cases were confirmed for the present V_2O_5 - Sb - TeO_2 glasses, because the decrease in W of the present glasses was not remarkable for $70V_2O_5 \cdot YSb \cdot (30 - Y)TeO_2$ glasses (mol%) ($Y = 5, 10$, and 20), similar to V_2O_5 - MnO - TeO_2 glasses [21]. Accordingly, so far as the present glasses are concerned, the hopping is considered to be mainly between V ions.

For $20V_2O_5 \cdot 20Sb \cdot 60TeO_2$ glass, however, the W value decreased (see Fig. 5), although the R value was the largest among the present glasses. This result was considered to be due to the influence of another electrical conduction or SPH mixed with another electrical conduction [15–18,22]. We then discuss the activation energy and the electrical conduction of $20V_2O_5 \cdot 20Sb \cdot 60TeO_2$ glass. The W values of $20V_2O_5 \cdot 20Sb \cdot 60TeO_2$ glasses were formed to be 0.412 eV , lesser than those of $30V_2O_5 \cdot 20Sb \cdot 50TeO_2$ and $40V_2O_5 \cdot 20Sb \cdot 40TeO_2$ glasses, although the conductivity of $20V_2O_5 \cdot 20Sb \cdot 60TeO_2$ glass was the lowest among the present glasses (Fig. 2). This lower W value was due to the fact that the slope in the conductivity-temperature relation for $20V_2O_5 \cdot 20Sb \cdot 60TeO_2$ glass was lesser than those for

$30V_2O_5 \cdot 20Sb \cdot 50TeO_2$ and $40V_2O_5 \cdot 20Sb \cdot 40TeO_2$ glasses as shown in Fig. 2. This result is the same as that for $Bi_2O_3 - SrO - V_2O_5$ glasses [22]. For $1.8Bi_2O_3 \cdot 16.8SrO \cdot 81.4V_2O_5$ glass, the slope in the conductivity-temperature relation ($T = 317$ – 330 K) indicated to be lower than that for $6.2Bi_2O_3 \cdot 9.3SrO \cdot 84.5V_2O_5$ glass, and the $1.8Bi_2O_3 \cdot 16.8SrO \cdot 81.4V_2O_5$ glass had a lower conductivity than the $6.2Bi_2O_3 \cdot 9.3SrO \cdot 84.5V_2O_5$ glass. Amano et al. [22] reported that this lower slope was considered to be due to a mixed conduction of SPH and variable-range hopping (VRH). Generally, the slope in the conductivity-temperature relation for VRH conduction is lower than that for SPH conduction, because VRH conduction exhibits a lower activation energy than SPH conduction [15–18,22]. The ternary glasses of the V_2O_5 - Sb_2O_3 - TeO_2 [17] and V_2O_5 - SnO - TeO_2 systems [18] indicated that a decrease of V_2O_5 concentration caused increases on the onset temperature of VRH conduction. This phenomenon was due to the fact that glass structure becomes more disordered with decreasing V_2O_5 concentration because the decrease appears in the form of a loosely packed structure [17,18], and hence the hopping electrons move between V-ions by using a lower activation energy compared with that for SPH conduction [17,18]. From this, the conductivity-temperature relation for VRH conduction indicates a lower slope. Therefore, we assume that the lower activation energy of $20V_2O_5 \cdot 20Sb \cdot 60TeO_2$ glass is affected by a mixed conduction of SPH and VRH or VRH conduction, because the concentration of V_2O_5 was the lowest among the present glasses; this result will be discussed in another paper in the near future.

4.3. Electrical conduction

The relationship between R and W (Fig. 5) suggested that small polaron hopping (SPH) between V-ions occurs in V_2O_5 - Sb - TeO_2 glasses. Generally, for glasses containing TMO [1–16], the electrical conduction at temperatures above about $\frac{1}{2}\theta_D$ is interpreted by the Austin-Mott theory of SPH conduction [38,39]. Thus, in this section, we discuss the applicability of SPH conduction based

on a single phonon approach to the present glasses in the high temperature region ($T > \frac{1}{2}\Theta_D$) and dependence of V_2O_5 -concentration on the conduction.

The conductivity (σ) for SPH in non-adiabatic region under strong electron–phonon interaction is expressed by the following equation [38,39].

$$\sigma = \frac{v_o N e^2 R^2}{kT} C(1 - C) \exp(-2\alpha R) \times \exp(-W/kT) \quad (6)$$

with

$$W = W_H + W_D/2 \quad (\text{for } T > \Theta_D/2), \quad (7a)$$

$$W = W_D \quad (\text{for } T < \Theta_D/4), \quad (7b)$$

where v_o is the optical phonon frequency (generally $v_o \approx 10^{13} \text{ s}^{-1}$ [13,21]), N the transition metal ion density, C the fraction of reduced transition metal ion ($= C_v$, and $C_v = V^{4+}/V_{\text{total}}$), R the average spacing between transition metal ions ($= (1/N)^{1/3}$), α the tunneling factor (the ratio of the wave function decay), W_H the hopping energy and W_D is the disorder energy defined as the difference of electronic energies between two hopping sites [40] and is given by [41]

$$W_D = (e^2/\epsilon_s R)K, \quad (7c)$$

where ϵ_s is the static dielectric constant, and K is a constant of order 0.3.

Then, σ_o in Eq. (1) is

$$\sigma_o = \frac{v_o N C(1 - C)e^2 R^2}{k} \exp(-2\alpha R). \quad (8)$$

In the adiabatic region, αR in Eq. (6) becomes negligible [11,42]. Since, $\exp(-2\alpha R) = 1$ [11,42], Eq. (6) is described in this region as follows:

$$\sigma = \frac{v_o N e^2 R^2}{kT} C(1 - C) \exp(-W/kT). \quad (9)$$

In order to confirm $\exp(-2\alpha R) = 1$ for the present glasses, the transition probability, $v_o \exp(-2\alpha R)$, in Eq. (6) was calculated using the experimental values in Tables 1 and 2. Fig. 6 shows the result, presenting $v_o \exp(-2\alpha R) \approx 10^{12} \text{ s}^{-1}$ for $V_2O_5 \geq 50 \text{ mol}\%$, which nearly agrees with reported data (10^{12} – 10^{13} s^{-1}) [1–5,13,14].

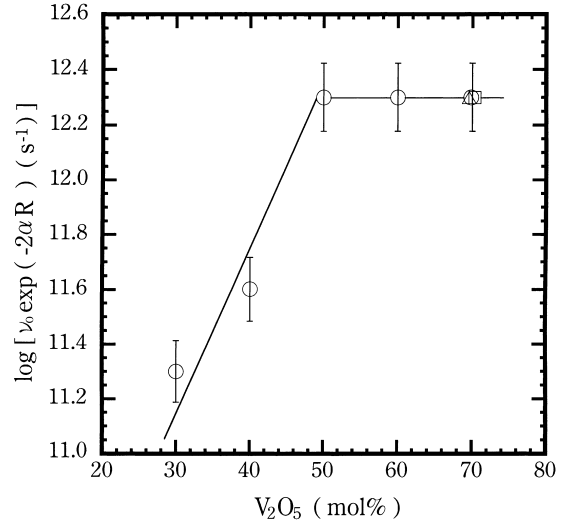


Fig. 6. Effect of V_2O_5 concentration on the transition probability, $v_o \exp(-2\alpha R)$. The line is drawn as a guide for the eye. (○) $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%), $X = 30$ – 70 . (△) $70V_2O_5 \cdot 5Sb \cdot 20TeO_2$ glass, (□) $70V_2O_5 \cdot 10Sb \cdot 20TeO_2$ glass.

Apart from this, we evaluate v_o in $v_o \exp(-2\alpha R)$ according to $k\Theta_D = h\nu_o$ (h is the Planck's constant) [5,13,14]. To determine v_o for the different compositions, the Debye temperature Θ_D was estimated by $T > \frac{1}{2}\Theta_D$ (Eq. (7a)) using the T values given in Table 1. Θ_D of the present glasses was obtained to be 666–806 K, which was nearly the same as the values of V_2O_5 – P_2O_5 [43] and alkaline silicate glasses [44]. Thus, these estimated Θ_D values indicate to be physically reasonable. Then, with the Θ_D values, v_o was calculated using $v_o = k\Theta_D/h$. The values of Θ_D and v_o are summarized in Table 1. For $V_2O_5 \geq 50 \text{ mol}\%$, the v_o and $v_o \exp(-2\alpha R)$ values agreed within one order, while for $30 \leq V_2O_5 < 50 \text{ mol}\%$, the v_o values indicated to be about two orders larger than the $v_o \exp(-2\alpha R)$ values. This result means that $\exp(-2\alpha R)$ can be regarded as nearly 1, i.e., $\alpha R \approx 0$ [42] for $V_2O_5 \geq 50 \text{ mol}\%$, similar to the previous vanadate glasses [1–5,11–14]. From these results, the conduction at $T > \frac{1}{2}\Theta_D$ for $V_2O_5 \geq 50 \text{ mol}\%$ was attributed to adiabatic SPH. On the other hand, for $30 \leq V_2O_5 < 50 \text{ mol}\%$, $v_o \exp(-2\alpha R)$ tended to decrease with decreasing V_2O_5 concentration in Fig. 6, which

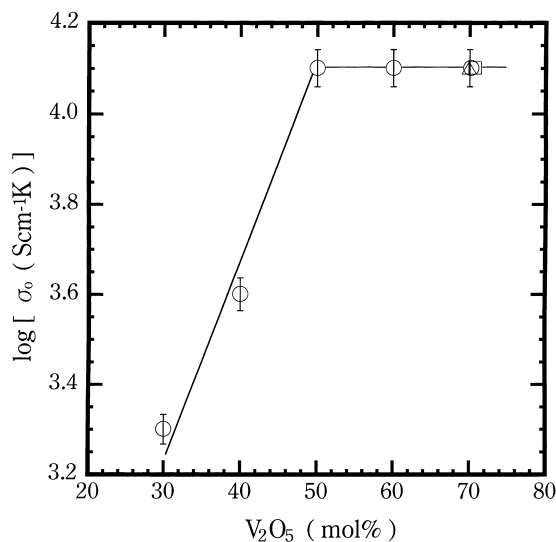


Fig. 7. Effect of V_2O_5 concentration on the pre-exponential factor, σ_0 . The line is drawn as a guide for the eye. (○) $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%), $X = 30$ –70. (△) $70V_2O_5 \cdot 5Sb \cdot 20TeO_2$ glass, (□) $70V_2O_5 \cdot 10Sb \cdot 20TeO_2$ glass.

indicated that the conduction was attributable to non-adiabatic SPH.

For adiabatic hopping conduction, the term of σ_0 in Eq. (1) was not dependent on V_2O_5 concentration, and hardly varied [1–5,11–14]. Therefore, the dominant factor contributing to the conductivity should be W in the adiabatic region [1–5, 11–14]. For the present glasses, we calculated the term of σ_0 using the experimental values in Tables 1 and 2. Fig. 7 shows the effect of V_2O_5 concentration on σ_0 . The σ_0 values hardly varied for $V_2O_5 \geq 50$ mol%, resulting that conduction is adiabatic SPH. In contrast, σ_0 decreased with decreasing V_2O_5 concentration for $30 \leq V_2O_5 < 50$ mol%, indicating the conduction to be non-adiabatic SPH. These results indicate that σ depends only on W in Eq. (9) in adiabatic region for the present glasses as well as in the earlier glasses [1–5,11–14]. Then, based on this result, $\log \sigma$ at a given temperature is proportional to W and the $\log \sigma - W$ relation should become $\log \sigma = \log \sigma_0 - W/2.303 kT$ from Eqs. (6) and (8). Fig. 8 shows the relationship between $\log \sigma$ at 473 K and W . This relationship was linear for $V_2O_5 \geq 50$ mol%. We fitted the data to the relation $\log \sigma \propto W$ for

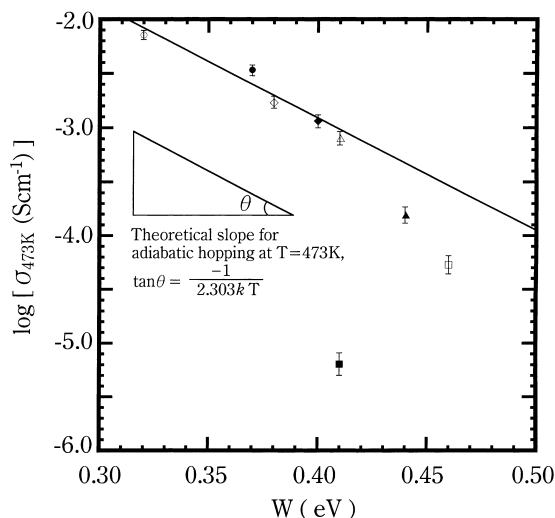


Fig. 8. Relationship between $\log \sigma$ at 473 K and activation energy, W , for V_2O_5 –Sb– TeO_2 glasses. $XV_2O_5 \cdot 0Sb \cdot m(80 - X)TeO_2$ glasses (mol%), (○) $X = 70$, (●) $X = 60$, (△) $X = 50$, (▲) $X = 40$, (□) $X = 30$, (■) $X = 20$. (◆) $70V_2O_5 \cdot 5Sb \cdot 20TeO_2$ glass, (◇) $70V_2O_5 \cdot 10Sb \cdot 20TeO_2$ glass. The lines are drawn by using the least-square technique for the W values between 0.32 and 0.41 eV.

$V_2O_5 \geq 50$ mol% by the least-square technique, and the slope of the regression line in Fig. 8 was obtained to be -10.55 eV^{-1} . This value was almost the same as the theoretical slope for the adiabatic hopping ($\tan \theta$ in Fig. 8), i.e. -10.65 eV^{-1} ($= -1/2.303 kT$, $T = 473 \text{ K}$). The equivalent temperature evaluated from the slope of the regression line gave $T = 477 \text{ K}$, which is nearly equal to the measured value $T = 473 \text{ K}$. On the other hand, the relationship for $30 \leq V_2O_5 < 50$ mol% deviated from the slope of the line. Thus, adiabatic conduction for $V_2O_5 \geq 50$ mol% was reconfirmed.

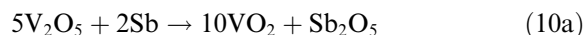
From these above discussions, we concluded that in high temperature region ($T > \frac{1}{2}\Theta_D$) the conduction for the present glasses was attributed to the adiabatic SPH of electrons for $V_2O_5 \geq 50$ mol%, and non-adiabatic SPH for $30 \leq V_2O_5 < 50$ mol%.

4.4. Fraction of reduced vanadium ion, C_v

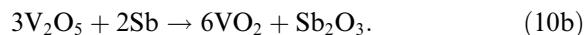
The experimental C_v values of V_2O_5 –Sb– TeO_2 glasses are given in Table 2. These became larger

than those of V_2O_5 – Sb_2O_3 – TeO_2 glasses (0.1–0.3) [1], suggesting a larger concentration of V^{4+} ions in these glasses compared to those for V_2O_5 – Sb_2O_3 – TeO_2 glasses [1]. Previously, we reported that the result of C_v for V_2O_5 – Sb_2O_3 – TeO_2 glasses obeyed the redox reaction $V_2O_5 + Sb_2O_3 \rightarrow 4VO_2 + Sb_2O_5$ during melting [1,45].

Then we discuss the redox reaction during melting for the V_2O_5 – Sb – TeO_2 glasses. Considering the reaction of V_2O_5 – Sb_2O_3 – TeO_2 glasses [1] and assuming different redox reaction models between V_2O_5 and Sb for the V_2O_5 – Sb – TeO_2 glasses, we express the reactions as follows:



or



The theoretical C_v values were calculated by using Eqs.(10a) and (10b), respectively, and the results (curves C_v -1(a) and (b)) and plots of the experimental C_v are shown in Fig. 9. As Fig. 9 indicates, the curves C_v -1(a) and (b) and the experimental C_v values did not agree. Thus, the redox reactions of

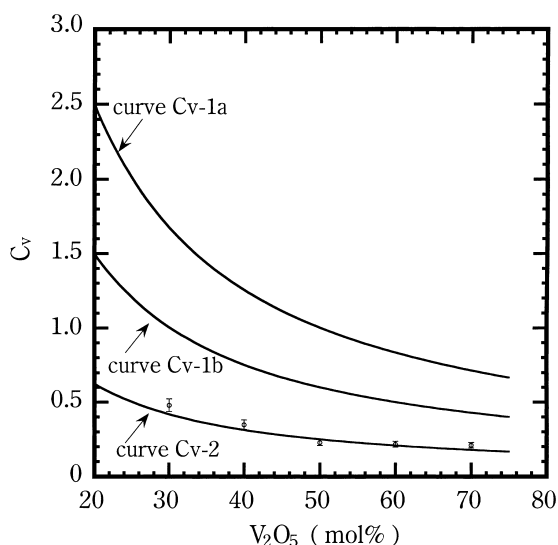
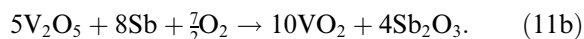


Fig. 9. Effect of V_2O_5 concentration on the fraction of reduced vanadium ion, C_v , for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%). The calculated curves C_v -1(a) and (b) and C_v -2 correspond to Eqs. (10a) and (10b), and Eqs. (11a) and (11b), respectively.

Eqs. (10a) and (10b) were inapplicable to explain the C_v data.

On the other hand, we discuss another redox reaction by considering that the present glasses were confirmed to have the closely packed structure by the reaction $TeO_4 \rightarrow TeO_3 + 1/2O_2$ (in Section 4.1). We assume the redox reaction expressed as follows:



Using Eqs. (11a) and (11b), the theoretical C_v values were calculated, and the result (curve C_v -2) is shown in Fig. 9. The agreement between the curve C_v -2 and the experimental C_v values was satisfactory. This agreement clearly indicates that the reaction expressed by Eq. (11a) affects the redox reaction (Eq. (11b)) between V_2O_5 and Sb, and the C_v values are determined by the reactions of Eqs. (11a) and (11b). From this, the antimony ion in the glasses was confirmed to be Sb^{3+} .

Next, we discuss the redox reaction of the present glasses on the basis of the V_O^* (see Section 4.1), d and C_v values. If the redox reaction of Eq. (10a), (10b) or Eqs. (11a) and (11b) is assumed during melting, the agreement between the experimental V_O^* values (plots in Fig. 4) and the theoretical value of V_O^* calculated using the theoretical C_v should be confirmed. To calculate the theoretical V_O^* value, an experimental density (d) – $[V_2O_5]$ relation was discussed by the regression analysis using the experimental d values (Table 1). Fig. 10 shows the relationship between d and V_2O_5 concentrations together with the curve drawn by the regression analysis. The regression curve indicates $d \text{ (g cm}^{-3}\text{)} = 4.8976 - 1.0411 \times 10^{-2} \times [V_2O_5 \text{ (mol\%)}] - 1.6626 \times 10^{-3} \times [V_2O_5 \text{ (mol\%)}]$. Then, using this regression equation and the curves C_v -1(a) and (b) in Fig. 9, we obtained the theoretical V_O^* curves as given in Fig. 4 (V_O^* -1(a) and (b), respectively). These curves in Fig. 4 were larger than the V_O^* values calculated using the experimental C_v and d values, and we confirmed that these curves did not agree with the experimental plots of V_O^* . On the other hand, the other theoretical V_O^* values were calculated using the curve

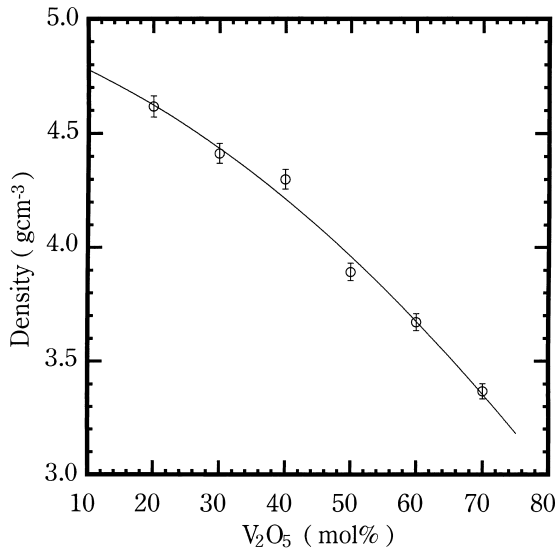


Fig. 10. Effect of V₂O₅ concentration on density, d , for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%). The curve is drawn using the regression analysis.

C_v-2 based on the redox model given by Eqs. (11a) and (11b). The result (curve $V_O^* - 2$) in Fig. 4 gives good agreements between the V_O^* plots and the curve V_O^*-2 , confirming again the redox reaction in Eqs. (11a) and (11b). Besides, this agreement also supports theoretically that the present glasses have the close packed structure.

4.5. Small polaron hopping parameters

SPH conduction at the temperatures above $\frac{1}{2}\Theta_D$ is discussed in terms of the polaron band width, J , expressed by the following equations [15,23,46,47]:

$$J > (2kT W_H/\pi)^{1/4} (h\nu_o/\pi)^{1/2} \quad (\text{adiabatic}), \quad (12)$$

$$J < (2kT W_H/\pi)^{1/4} (h\nu_o/\pi)^{1/2} \quad (\text{non-adiabatic}). \quad (13)$$

For adiabatic hopping conduction, the hopping energy, W_H , is given using J as

$$W - W_D/2 = W_H = W_p/2 = W'_p/2 - J, \quad (14)$$

where W_p is the polaron binding energy, W'_p is the maximum polaron binding energy and W_H depends

on R [13]. By contrast, for non-adiabatic hopping conduction, W_H is given as follows:

$$W - W_D/2 = W_H = W'_p/2. \quad (15)$$

In order to determine the W_H value of the present glasses, we evaluated the disorder energy (the difference of electronic energies between two hopping sites [40]), W_D , from the measurement of low-temperature conductivity. The W_D values were obtained to be 0.030 ± 0.004 eV for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 50-70$) (the result of the conductivities and the electrical conduction at lower temperatures ($T \lesssim RT$) for the present glasses will be discussed elsewhere).

These W_D values indicated to be of the same order, compared with those for V_2O_5 - TeO_2 glasses [15] and V_2O_5 - R - TeO_2 ($R = Sb_2O_3$ [17] or SnO [18]), i.e., $W_D = 0.05-0.09$ eV. With $W_D = 0.030$ eV, the W_H values of the present glasses were calculated from Eq. (7a) using W at 423–473 K. The result indicated to be $W_H(423-473K) = 0.302-0.450$ eV for $V_2O_5 = 30-70$ mol% (Table 2).

For $V_2O_5 < 50$ mol%, W_H is nearly equal to $W'_p/2$ [13]. Thus, from the W_H values of the present glasses, the maximum value of $W'_p/2$ was obtained to be ~ 0.450 eV. Then, the J values were estimated to be $0.022-0.148$ eV by using W_H , $W'_p/2$ and Eq. (14). Both W_H and J values at 423–473 K are given in Table 2.

It is noted that the following criterion [48] should be satisfied for SPH conduction:

$$J < 1/3(W'_p/2). \quad (16)$$

The J values of the present glasses were obtained to be ≤ 0.148 eV, hence the above criterion was satisfied because the right-hand term in Eq. (16) was 0.150 eV. The right-hand terms in Eqs. (12) and (13) were calculated to be 0.035 ± 0.002 eV at 423–473 K for $h\nu_o = 0.3$ eV and $\nu_o = 10^{13}$ s⁻¹ as well as in the systems V_2O_5 - R - TeO_2 ($R = Sb_2O_3$ [1], SnO [2], Bi_2O_3 [3] or ZnO [4]) and PbO - P_2O_5 - V_2O_5 [13]. In the adiabatic region for the present glasses, the criteria (Eqs. (12) and (16)) were satisfied, because $J = 0.052-0.148$ eV. On the other hand, in the non-adiabatic region, the J value was found as < 0.037 eV from Eq. (13) in the conduction for non-adiabatic hopping. From the

evaluated J values, we reconfirmed adiabatic SPH for $V_2O_5 \geq 50$ mol%, and non-adiabatic SPH for $40 \leq V_2O_5 < 50$ mol%. However, it should be noted that the J value was not estimated for $V_2O_5 = 30$ mol%, because the maximum value of $W'_p/2$ is the hopping energy of the $30V_2O_5 \cdot 20Sb \cdot 50TeO_2$ glass (sample no. 7). In addition, in order to calculate hopping carrier mobility and density (see Section 4.6) of the $30V_2O_5 \cdot 20Sb \cdot 50TeO_2$ glass, the J value should be determined. Then, in the present study, the J value was assumed to be 0.020 eV, because the electrical conduction of the $30V_2O_5 \cdot 20Sb \cdot 50TeO_2$ glass (mol%) was concluded to be the non-adiabatic SPH from the discussion in the Section 4.3.

Next, we discuss the polaron hopping parameters using the values of R , W_H and J . The polaron radius r_p is given by [49]

$$r_p = (\pi/6)^{1/3} (R/2). \quad (17)$$

By using Eq. (17), r_p was obtained to be $r_p = 0.154\text{--}0.187$ nm (Table 2). The polaron hopping energy, W_H , is expressed by [39]

$$W_H = (e^2/4\epsilon_p)(r_p^{-1} - R^{-1}), \quad (18)$$

where $\epsilon_p^{-1} = \epsilon_\infty^{-1} - \epsilon_0^{-1}$, ϵ_0 and ϵ_∞ are the static and high-frequency dielectric constants of glass, respectively, and ϵ_p is the effective dielectric constant. With W_H , r_p and R values, we obtained $\epsilon_p = 1.80\text{--}3.12$ for the present glasses. These values were smaller than those of $V_2O_5\text{--}Bi_2O_3$ doped with $BaTiO_3$ [19].

J is expressed in terms of the density of states at the Fermi level $N(E_F)$ as follows [15]:

$$J = e^3 N(E_F)^{1/2} / \epsilon_p^{3/2}. \quad (19)$$

From the result of the calculation by using Eq. (19), $N(E_F)$ was estimated to be $2.32 \times 10^{18}\text{--}5.94 \times 10^{20}$ eV⁻¹ cm⁻³ at 423–473 K, as given in Table 2 together with the ϵ_p data. The values of $N(E_F)$ were nearly the same as those of $V_2O_5\text{--}MnO\text{--}TeO_2$ glasses obtained from the analysis of SPH ($2.4 \times 10^{17}\text{--}2.9 \times 10^{18}$ eV⁻¹ cm⁻³ at 405 K [21]), while the values indicated to be lower than those of $V_2O_5\text{--}Sb_2O_3\text{--}TeO_2$ glasses ($1.26 \times 10^{21}\text{--}9.30 \times 10^{22}$ eV⁻¹ cm⁻³ at 250 K [17]) evaluated by using the model of variable-range hopping.

The values of small polaron coupling constant (γ_p), which is a measure of the electron–phonon interaction, were also evaluated for the present glasses using the formula $\gamma_p = 2W_H/h\nu_0$ [50]. The values of γ_p were 10.6–12.8 (Table 2), indicating to be larger than those for $V_2O_5\text{--}Bi_2O_3$ doped with $BaTiO_3$ (7.05–7.60) [19]. Such large values were of the same order as those of $V_2O_5\text{--}Sb_2O_3\text{--}TeO_2$ glasses (18–24) [1], $V_2O_5\text{--}MnO\text{--}TeO_2$ glasses (21–26) [21] and $V_2O_5\text{--}NiO\text{--}TeO_2$ glasses (19–23) [24]. Thus, we concluded a strong electron–phonon interaction for the present glasses, similar to $V_2O_5\text{--}Sb_2O_3\text{--}TeO_2$ [1], $V_2O_5\text{--}MnO\text{--}TeO_2$ [21] and $V_2O_5\text{--}NiO\text{--}TeO_2$ glasses [24].

4.6. Hopping carrier mobility and density

The hopping carrier mobility, μ , in the adiabatic and non-adiabatic hopping regions is described by the following equations [38,51]:

$$\mu = (v_0 e R^2 / kT) \exp(-W_H / kT) \quad (\text{adiabatic}), \quad (20)$$

$$\mu = (e R^2 / kT) (1/\hbar) (\pi / 4 W_H kT)^{1/2} J^2 \exp(-W / kT) \quad (\text{non-adiabatic}). \quad (21)$$

The μ values of the present glasses were calculated using the experimental data in Tables 1 and 2, and the result is given in Table 3. The μ values of adiabatic SPH conduction varied from 3.78×10^{-5} to 2.90×10^{-4} cm² V⁻¹ s⁻¹ at 473 K, and for non-adiabatic SPH conduction, the values were $(2.65\text{--}3.75) \times 10^{-6}$ cm² V⁻¹ s⁻¹ at 473 K. From these results, we confirmed that the μ values show that $\mu \ll 10^{-2}$ cm² V⁻¹ s⁻¹ which is the requirement for hopping conduction [52]. Fig. 11 shows the effect of V_2O_5 concentration on μ for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%) ($X = 30\text{--}70$) at 423 and 473 K. The μ values increased with increasing V_2O_5 concentration in adiabatic and non-adiabatic regions. In this figure, the experimental relation $\mu \propto \exp[V_2O_5]$ was exhibited. However, the change in the μ value was discontinuous, when the conduction changed from adiabatic to non-adiabatic SPH. This result was also observed for the system $V_2O_5 - R - TeO_2$

Table 3

Hopping carrier mobility and density of V_2O_5 -Sb- TeO_2 glasses

Sample no.	μ ($cm^2 V^{-1} s^{-1}$) (473 K)	N_c (cm^{-3}) (423–473 K)
1	$(4.55 \pm 1.28) \times 10^{-5}$	$(8.62 \pm 4.00) \times 10^{20}$ – $(9.00 \pm 3.00) \times 10^{20}$
2	$(7.08 \pm 1.00) \times 10^{-5}$	$(1.41 \pm 0.50) \times 10^{21}$ – $(1.47 \pm 0.12) \times 10^{21}$
3	$(2.90 \pm 0.50) \times 10^{-4}$	$(1.86 \pm 0.20) \times 10^{21}$ – $(1.94 \pm 0.20) \times 10^{21}$
4	$(9.09 \pm 2.00) \times 10^{-5}$	$(1.81 \pm 0.15) \times 10^{21}$ – $(1.89 \pm 0.10) \times 10^{21}$
5	$(3.78 \pm 0.60) \times 10^{-5}$	$(1.68 \pm 0.10) \times 10^{21}$ – $(1.76 \pm 0.10) \times 10^{21}$
6	$(3.75 \pm 1.50) \times 10^{-6}$	$(3.11 \pm 0.40) \times 10^{20}$ – $(3.29 \pm 0.50) \times 10^{20}$
7	$(2.65 \pm 0.50) \times 10^{-6}$	$(1.35 \pm 1.20) \times 10^{20}$ – $(1.43 \pm 1.20) \times 10^{20}$

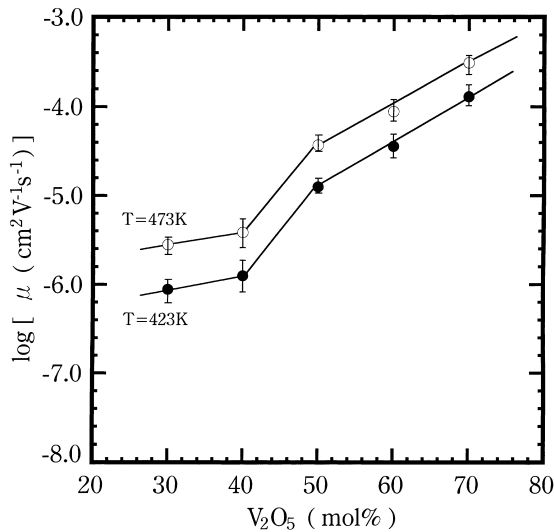


Fig. 11. Effect of V_2O_5 concentration on hopping carrier mobility, μ , for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%). The lines are drawn as a guide for the eye.

($R = Sb_2O_3$ [1], SnO [2], Bi_2O_3 [3] or ZnO [4]). This fact was interpreted as a result of the influence of the J term in Eq. (21) for the non-adiabatic case [1–4].

Next, we discuss carrier density, N_c , given as follows [51]:

$$\sigma = N_c e \mu. \quad (22)$$

Then, N_c , i.e., $\sigma/e\mu$ can be expressed by Eqs. (23) and (24), combined with Eqs. (6), (20) and (21),

$$N_c = NC(1 - C) \exp(-W_D/2kT) \quad (\text{adiabatic}), \quad (23)$$

$$N_c = \frac{v_o \exp(-2\alpha R) NC(1 - C)}{(1/\hbar)(\pi/4W_H kT)^{1/2} J^2} \quad (\text{non-adiabatic}). \quad (24)$$

N_c was calculated using the data in Tables 1 and 2. The N_c values at 423–473 K are also given in Table 3. From the results of μ and N_c , it was found that σ of the present glasses was determined by μ , because the N_c values were almost unchanged at the temperatures between 423 and 473 K.

For the present glasses, the antimony ions in the glasses were confirmed to be Sb^{3+} from the discussion on the C_v values in Section 4.4. The ionic radius of Sb^{3+} (0.077 nm) is larger than that of Sb^{5+} (0.061 nm) [32]. However, the present glasses indicated that the glasses had larger conductivities than those of V_2O_5 - Sb_2O_3 - TeO_2 glasses with the ionic radius of Sb^{5+} . Thus, regarding this result, we discuss the conductivity from view points of the C_v , μ and N_c . For the $62V_2O_5 \cdot 4Sb_2O_3 \cdot 33TeO_2$ glass, μ was estimated to be $4.6 \times 10^{-5} cm^2 V^{-1} s^{-1}$ at 473 K [1] using $R = 0.4061 nm$ and $W = 0.38 eV$. From this, it was confirmed that $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%)($X = 60$ and 70) had higher μ values (9.09×10^{-5} and $2.90 \times 10^{-4} cm^2 V^{-1} s^{-1}$ at 473 K) than the $62V_2O_5 \cdot 4Sb_2O_3 \cdot 33TeO_2$ glass. In addition, the N_c values at 423–473 K for $XV_2O_5 \cdot 20Sb \cdot (80 - X)TeO_2$ glasses (mol%)($X = 60$ and 70) were obtained to be $(1.81\text{--}1.89) \times 10^{21} cm^{-3}$ and $(1.86\text{--}1.94) \times 10^{21} cm^{-3}$, respectively, indicating that they were larger than that for the $62V_2O_5 \cdot 4Sb_2O_3 \cdot 33TeO_2$ glass ($1.37 \times 10^{21} cm^{-3}$ at 340–473 K [1]). These larger N_c values of the glasses (mol%)($X = 60$ and 70) were due to the fact that the glasses had a larger C_v value than the $62V_2O_5 \cdot 4Sb_2O_3 \cdot 33TeO_2$ glass, because a larger concentration of V^{4+} ions for the present glasses ($X = 60$ and 70) was obtained compared to that for the $62V_2O_5 \cdot 4Sb_2O_3 \cdot 33TeO_2$ glass (V^{4+} ion = $3.48 \times 10^{21} cm^{-3}$ for $X = 60$ mol% and V^{4+} ion =

$3.56 \times 10^{21} \text{ cm}^{-3}$ for $X = 70 \text{ mol\%}$, respectively, while V^{4+} ion $= 2.49 \times 10^{21} \text{ cm}^{-3}$ for the $62\text{V}_2\text{O}_5 \cdot 4\text{Sb}_2\text{O}_3 \cdot 33\text{TeO}_2$ glass [1]). Thus, both these larger μ and N_c values explain the result indicating the high conductivities of the glasses compared to that of the $62\text{V}_2\text{O}_5 \cdot 4\text{Sb}_2\text{O}_3 \cdot 33\text{TeO}_2$ glass.

Next, the σ values of the $70\text{V}_2\text{O}_5 \cdot Y\text{Sb} \cdot (30 - Y)\text{TeO}_2$ glasses (mol%) ($Y = 5, 10, \text{ and } 20$) are discussed from view points of μ and N_c . For these glasses, the μ values at 473 K were obtained to be 4.55×10^{-5} – $2.90 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ (Table 3), indicating that μ increased with increasing Sb concentration. In addition, N_c also increased with increasing Sb concentration. The increase of N_c was attributed to an increase of concentration of V^{4+} ions in $C_v = \text{V}^{4+}/\text{V}_{\text{total}}$. Therefore, it was concluded for the $70\text{V}_2\text{O}_5 \cdot Y\text{Sb} \cdot (30 - Y)\text{TeO}_2$ glasses (mol%) ($Y = 5, 10, \text{ and } 20$) that the reason for the increase of σ was due to the increases of μ and N_c with increasing Sb concentration. Particularly, for these glasses, μ remarkably increased with increasing Sb concentration, e.g., the μ value of the $70\text{V}_2\text{O}_5 \cdot 20\text{Sb} \cdot 10\text{TeO}_2$ glass was about nine times greater than that of the $70\text{V}_2\text{O}_5 \cdot 5\text{Sb} \cdot 25\text{TeO}_2$ glass.

Regarding the effect of Sb content on μ , we discuss μ from viewpoints of W and ionic radius of Sb, because W depends on R and ionic radius in SPH conduction [1–4]. Hirashima et al. [13] reported that the distance between V-ions of linear bonding, V–O–V of the system $\text{PbO–P}_2\text{O}_5\text{–V}_2\text{O}_5$, was estimated to be about 0.39 nm by using the ionic radii of O^{2-} ($= 0.14 \text{ nm}$) [32] and V^{5+} ($= 0.054 \text{ nm}$) [32] in crystalline oxides. On the other hand, the ionic radii of V^{4+} , Sb^{3+} and Te^{6+} are 0.059, 0.077 and 0.056 nm, respectively [32]. Therefore, the largest ionic radius is O^{2-} , and hence, O ions in a glass act as a large obstacle for the hopping electrons. Then, we discuss the change in the O-ion and Sb-ion densities ($N_{\text{O-ion}}$ and $N_{\text{Sb-ion}}$) when Sb concentration increases, by calculating O-ion and Sb-ion densities for the $70\text{V}_2\text{O}_5 \cdot Y\text{Sb} \cdot (30 - Y)\text{TeO}_2$ glasses (mol%) ($Y = 5, 10, \text{ and } 20$). The O-ion and Sb-ion densities can be calculated using the following equations.

$$N_{\text{O-ion}} = 5[(d W_{\text{V}_2\text{O}_5}/MW_{\text{V}_2\text{O}_5})N_A] + 2[(d W_{\text{TeO}_2}/MW_{\text{TeO}_2})N_A], \quad (25)$$

$$N_{\text{Sb-ion}} = (d W_{\text{Sb}}/MW_{\text{Sb}})N_A, \quad (26)$$

where W_{TeO_2} is the weight fraction of TeO_2 , W_{Sb} the weight fraction of Sb, MW_{TeO_2} the molecular weight of TeO_2 and MW_{Sb} is the molecular weight of Sb, respectively. Fig. 12 shows the effects of Sb concentration on O-ion and Sb-ion densities.

From this figure, it was found that $N_{\text{O-ion}}$ decreased, while $N_{\text{Sb-ion}}$ increased, with increasing Sb concentration. This result indicated that the O-ions in the hopping path of electron were substituted by Sb-ions having a lower ionic radius than that of O-ions. On the other hand, it should be noted that N (Table 1) was almost unchanged for these glasses. Thus, these results mean that the hopping electrons become able to move easily between V-ions with lowering activation energy with increasing Sb concentration (see sample 1–3 in Table 1).

Therefore, it was concluded that the change in $N_{\text{O-ion}}$, which is large obstacle in the hopping path, affected the term of W_H in μ , and that the conductivity of $70\text{V}_2\text{O}_5 \cdot Y\text{Sb} \cdot (30 - Y)\text{TeO}_2$ glasses (mol%) ($Y = 5, 10, \text{ and } 20$) was determined almost uniquely by the μ term in Eqs. (20) and (22).

Finally, the electrical conduction at $T < \frac{1}{2}\Theta_D$ for the present glasses is discussed on the basis of

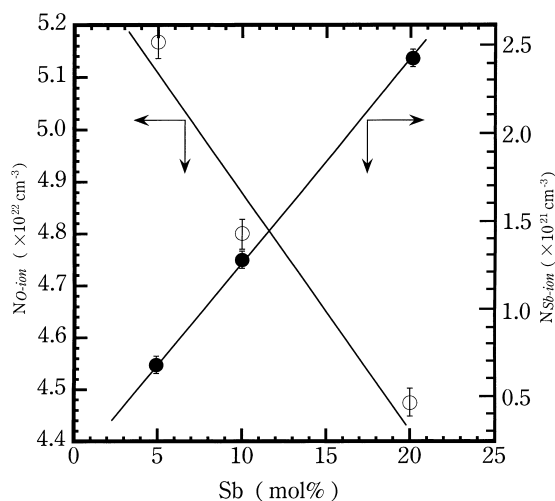


Fig. 12. Effect of Sb concentration on O-ion and Sb-ion densities for $70\text{V}_2\text{O}_5 \cdot Y\text{Sb} \cdot (30 - Y)\text{TeO}_2$ glasses (mol%). (○) $N_{\text{O-ion}}$, (●) $N_{\text{Sb-ion}}$. The lines are drawn as a guide for the eye.

Schnakenberg's model [25], because of the deviation of the $\log(\sigma T) - T^{-1}$ relation from Eq. (6). Schnakenberg [25] has suggested that when the temperature is lowered, the multiphonon processes are replaced by a single optical phonon process, and that the polarons hop with one or more acoustic phonons making up the differences between sites at the lowest temperatures. The reduction of activation energy with temperature in the intermediate temperature region is expressed by the following relation [25]:

$$W = W' \frac{\tan h[(1/4)\hbar\omega_{\text{ph}}\beta]}{(1/4)\hbar\omega_{\text{ph}}\beta}, \quad (27)$$

where W and W' are the low- and high-temperature activation energies, respectively, $\beta = 1/kT$, $\omega_{\text{ph}} = 2\pi\nu_0$ and $\hbar = h/2\pi$.

Since intermediate-temperature ($T < \frac{1}{2}\Theta_D$) conductivities were measured for $X\text{V}_2\text{O}_5 \cdot 20\text{Sb} \cdot (80 - X)\text{TeO}_2$ glasses (mol%) ($X = 50$ –70), the experimental W/W' values of the glasses can be plotted against $1/T$. For example, the $70\text{V}_2\text{O}_5 \cdot 20\text{Sb} \cdot 10\text{TeO}_2$ glass is plotted in Fig. 13 together with the theoretical values calculated from Eq. (27). The experimental value of the $70\text{V}_2\text{O}_5 \cdot 20\text{Sb} \cdot 10\text{TeO}_2$ glass decreased with decreasing temperature, but the quantitative fit with the theoretical plot of Eq. (27) were rather poor ($X\text{V}_2\text{O}_5 \cdot 20\text{Sb} \cdot (80 - X)\text{TeO}_2$ glasses (mol%) ($X = 50$ and 60) were also confirmed to be poor, although the experimental W/W' values decreased with decreasing temperature). These results indicate that the decrease in the values of conductivity with temperature cannot be accounted for by the decrease in the activation energy alone (i.e., the effect of acoustic phonon). Thus, Schnakenberg's model for polaronic conduction can qualitatively explain the present conductivity data at $T < \frac{1}{2}\Theta_D$, but fail to give any quantitative agreement.

Further, the possibilities of applications of Mott's VRH model [53] and Greave's intermediate-temperature VRH model [54] to the transport of the present glasses in the intermediate temperature region should also be discussed, although the conduction in this temperature region can be qualitatively explained by using Schnakenberg's model [25]. We shall discuss the conduction

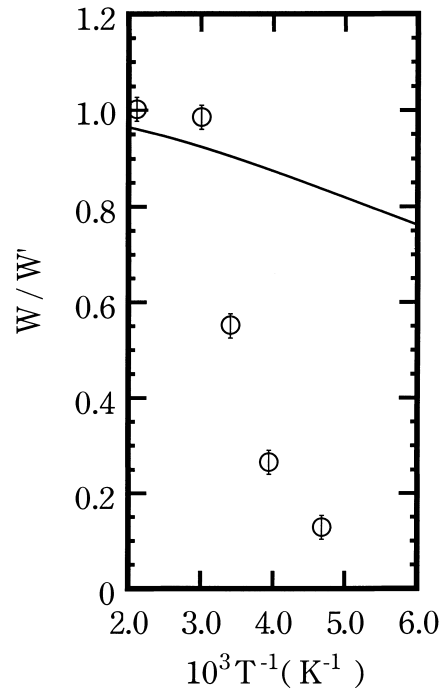


Fig. 13. Plot of W/W' as a function of T^{-1} for the $70\text{V}_2\text{O}_5 \cdot 20\text{Sb} \cdot 10\text{TeO}_2$ glass (mol%). The line is calculated using Eq. (27).

mechanism in the lower temperature region elsewhere in the near future.

5. Conclusions

Semiconducting glasses in the system V_2O_5 –Sb– TeO_2 were prepared by press quenching method and the electrical conduction mechanism was studied in terms of temperature dependence of dc conductivity, σ . The glasses were n-type semiconducting, and the composition range of glass formation was $10 \leq \text{TeO}_2 \leq 100$ mol%, $0 < \text{V}_2\text{O}_5 \leq 70$ mol%, and $0 < \text{Sb} \leq 20$ mol%, respectively. The conductivity at 473 K ranged from 6.38×10^{-6} to $7.13 \times 10^{-3} \text{ S cm}^{-1}$. σ of the present glasses were greater than those of the V_2O_5 – Sb_2O_3 – TeO_2 glasses. A redox reaction model of the present system was proposed, which explained quantitatively the experimental values of the fraction of vanadium ion (C_v) and the oxygen molar volume (V_O^*) as well. Thus, Sb element acted

as an effective reducing element to increase V^{4+} ion concentration.

The electrical conduction in high temperature region (temperature higher than half of the Debye temperature, Θ_D) was confirmed to be due to adiabatic small polaron hopping of electrons between vanadium ions for $V_2O_5 \geq 50$ mol%, and non-adiabatic for $30 \leq V_2O_5 < 50$ mol%. This corresponded to relatively large small polaron coupling constants ($\gamma_p = 10.6$ –12.8).

The polaron band width (J) was estimated to be $J = 0.022$ – 0.148 eV, and the J values indicated the conduction to be due to small polaron hopping. The polaron hopping carrier mobility (μ) was from 2.65×10^{-6} to 2.90×10^{-4} $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$ at 473 K. The carrier density (N_c) was obtained to be the order of 10^{20} – 10^{21}cm^{-3} , and the N_c values were almost unchanged at temperatures between 423 and 473 K. From this, it was found that the dominant factor determining conductivity was the hopping carrier mobility.

At temperatures lower than $\frac{1}{2}\Theta_D$, the conductivity deviated appreciably from the conductivity–temperature relation, which suggested the effect of acoustic phonons or variable-range hopping. We shall discuss the conduction in the intermediate temperature region and at the temperature below RT elsewhere in the near future.

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